



SAFER STREETS PRIORITY FINDER VERSION 2.0

A Tool and Dashboard for Safety
Analysis

Version 2.0 Report

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Project and Report Details

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1. INTRODUCTION

Traffic safety continues to be a major public health issue in the United States. In 2024 alone, there were over 39,000 traffic fatalities and an estimated 2.42 million people injured in traffic crashes¹. While the fatality number has declined from its 2020 peak and has shown a downward trend, it remains higher than it was a decade ago². To reverse this trend, agencies need practical tools that help them understand the safety conditions of their roadway networks and identify where the greatest risks lie. Targeting safety investments effectively depends on being able to pinpoint not only where crashes have occurred, but where there is a crash risk present.

Most agencies that prioritize safety investments rely on observed crash history, using proven approaches such as crash summaries, hotspot analyses, or High Injury Networks built from recent crash records. These methods are valuable and widely used: they reliably identify where severe crashes have already occurred and help agencies direct resources to demonstrated problem areas. At the same time, because they are built on recorded crashes, they can leave gaps, potentially missing locations where the conditions for severe crashes exist but crashes have not yet been recorded. These gaps tend to be most pronounced for pedestrian and bicyclist crashes, which are statistically rare on any given segment and unevenly reported across jurisdictions.

The Safer Streets Priority Finder (SSPF) was originally created to lower the technical barrier to safety analyses, including crash summaries, High Injury Networks, and crash risk analysis. Version 1.0³, released in 2021 with support from the USDOT Safety Data Initiative, combined a national pedestrian fatality model⁴, sliding-window analysis, and publicly available data into a single open-source web application focused on vulnerable road users. Version 2.0⁵ extends that framework to pedestrian, bicyclist, and motor vehicle modes across the full KABCO severity scale, with substantially expanded user control and a more stable computational core.

This report is written for the practitioners who will use the tool: planners and engineers at municipal, county, regional, state agencies, and private sector who need to identify candidate corridors for safety investment, but who may not have the GIS or statistical infrastructure to develop a full systemic analysis from scratch. The remainder of the report explains what is new in Version 2.0, how the tool works, what data it requires, how to configure and interpret an analysis, how the underlying models were validated, and where the tool is and is not appropriate to use.

¹ National Center for Statistics and Analysis. (2026, April). Overview of motor vehicle traffic crashes in 2024 (Traffic Safety Facts Research Note. Report No. DOT HS 813 791). National Highway Traffic Safety Administration. <https://doi.org/10.21949/grds-1877>

² Addressing America's Traffic Safety Crisis: Examining the Causes of Increasing U.S. Traffic Fatalities and Identifying Solutions to Improve Traffic Safety - https://tripnet.org/wp-content/uploads/2025/07/TRIP_Addressing_Americas_Traffic_Safety_Crisis_Report_July_2025.pdf

³ SSPF v1.0 Report - https://sspf.tooledesign.com/assets/documents/SSPF%20v1.0%20final_report.pdf

⁴ Mansfield, T. J., Peck, D., Morgan, D., McCann, B., & Teicher, P. (2018). The effects of roadway and built environment characteristics on pedestrian fatality risk: A national assessment at the neighborhood scale. *Accident; analysis and prevention*, 121, 166–176. <https://doi.org/10.1016/j.aap.2018.06.018>

⁵ SSPF v2.0 Website - <https://sspf.tooledesign.com/>

2. WHAT'S NEW IN VERSION 2.0

Version 2.0 was shaped by several years of experience using Version 1.0 across agencies of different sizes, geographies, and analytical capacities. Four themes drove the update: **broader modal coverage, alignment with current safety research practice, faster and more stable analysis runs, and greater user control over the parameters that drive results.**

2.1 EXPANDING MODAL COVERAGE

Version 1.0 was built around the USDOT Pedestrian Fatality Model and was applied post-hoc to bicyclist analysis. Version 2.0 replaces that single tract-level model with a suite of national block-group fatal crash prediction models covering pedestrian, bicyclist, and motor vehicle modes separately. The same Bayesian framework is then applied across all three modes and across the full KABCO severity scale (K, A, B, C, and O), allowing motorist risk to be screened alongside vulnerable road user risk in a consistent way.

2.2 UPDATED BAYESIAN MODEL

The Bayesian update procedure in Version 2.0 has been redesigned to align more closely with empirical Bayes methods documented in the Highway Safety Manual⁶ and refined in subsequent safety research. Segment-level priors are derived from the national block-group models by way of a transparent disaggregation step, observed crashes are redistributed spatially to reflect the imprecision of recorded crash locations, and the prior-versus-observed blend is governed by the coefficient of variation of the underlying block-group estimate.

2.3 FASTER, MORE STABLE ANALYSIS

Version 1.0 used Markov chain Monte Carlo (MCMC) simulation through RStan, which produced statistically rigorous estimates but often required several hours to complete an analysis and was prone to convergence issues on smaller study areas. Version 2.0 replaces the MCMC engine with a deterministic empirical Bayes update applied to pre-computed priors - national block-group level fatal crash estimates, state level crash rates by mode and functional class, and state level crash severity ratios by mode. Analyses that previously took hours now run in minutes and yield an output within the user's current browser session.

⁶ American Association of State Highway and Transportation Officials. (2010). Highway Safety Manual (1st ed.). AASHTO.

2.4 ANALYSIS CUSTOMIZATION

Version 1.0 allowed users to adjust only the crash cost values. Version 2.0 makes additional parameters configurable, including the severity weights used in the sliding windows analysis, the sliding window length and short-window ratio, the crash costs by severity level, the crash-to-segment join distance, and the number of top corridors reported. Each parameter has a sensible default; the defaults can be overridden, however, where local policy or local crash characteristics warrant. Section 5 describes the full set of configurable parameters and their allowed ranges.

3. ANALYTICAL METHODS

SSPF v2.0 produces two independent analyses of the same roadway network: a crash-history based analysis built from the Sliding Windows Analysis and a model-based analysis built from the Safer Streets Model. The two analyses are complementary, and their outputs are at a network segment level which can be used in planning processes (Figure 1). Sliding windows analysis answers the question "where have severe or high concentration of crashes already occurred?" The Safer Streets Model answers the question "where are the conditions for severe crashes present, regardless of whether crashes have been recorded yet?" Practitioners typically use both screening methods together - corridors that score high on both signal strong investment candidates; corridors that score high only on the model signal systemic opportunities that conventional analysis would miss.

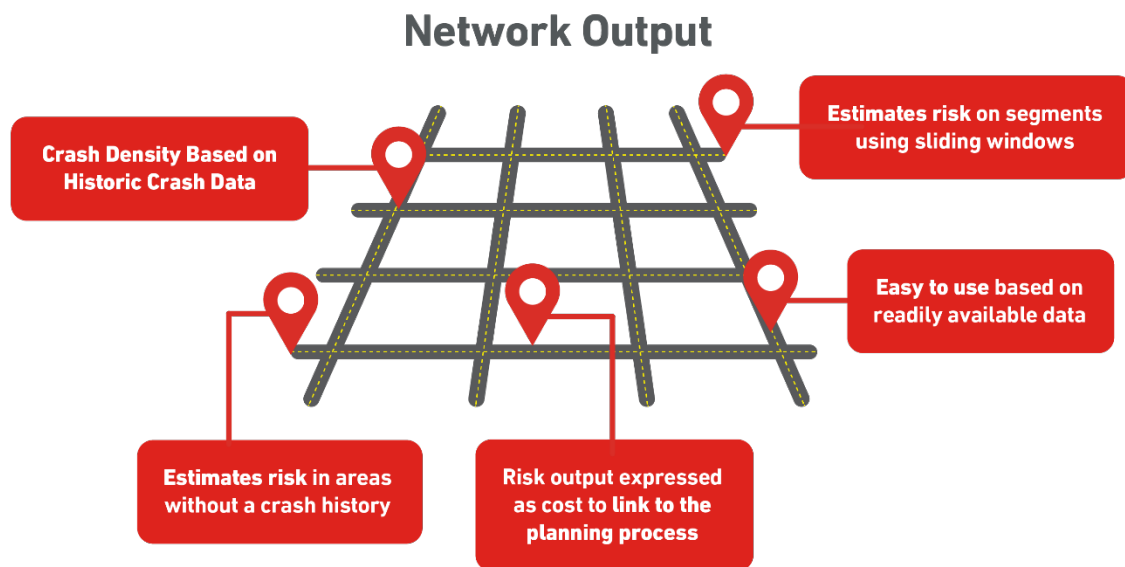


Figure 1: SSPF v2.0 analyses outputs are at a network segment level that can directly be used in planning processes.

3.1 SLIDING WINDOWS ANALYSIS

The Sliding Windows Analysis is a traditional crash-history based approach to identifying high-crash corridors. A fixed-length window is moved along each corridor in small incremental steps. At each step, the crashes inside the window are summed, weighted by KABCO severity, and assigned to a short underlying segment so that overlapping windows can be compared at a consistent scale. The result is a severity-weighted crash density score for every short segment in the network, by mode. The process is illustrated in Figure 2.

Sliding Window Analysis

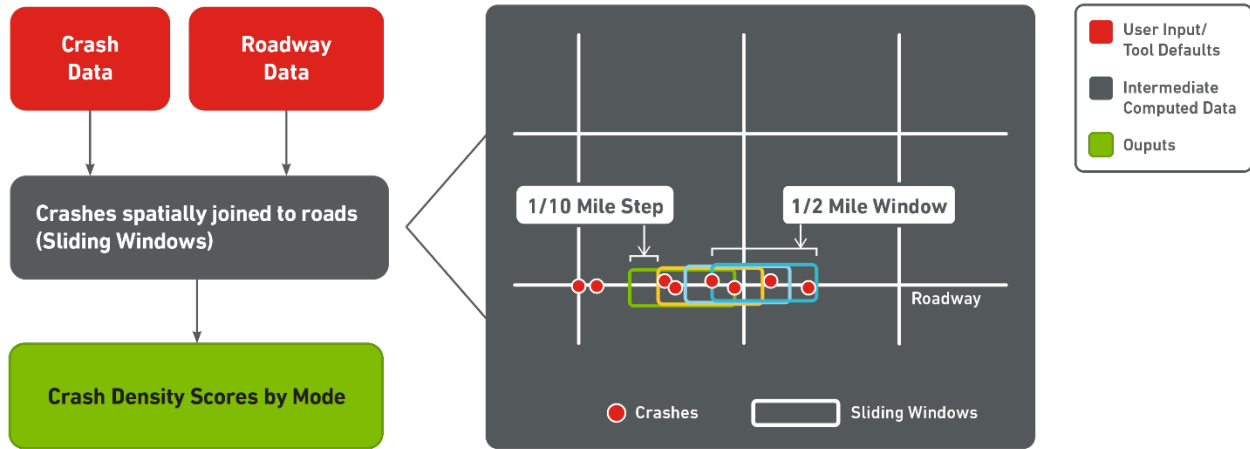


Figure 2: Sliding Windows Analysis Process

Corridors with elevated scores reflect both the frequency and the severity of recent crashes. Sliding windows analysis is fast to compute, easy to communicate, and grounded in directly observed data. It is, however, only as reliable as the underlying crash record - corridors with sparse or under-reported crash histories will score low even when the conditions for severe crashes are present.

3.2 SAFER STREETS MODEL

The Safer Streets Model estimates segment-level crash risk by combining a model-based prior (an initial, model-based estimate that represents what we expect before any locally observed data is considered) with locally observed crash history through a Bayesian update. The prior draws on several model-based inputs: 1. a national Block Group fatal crash risk model, which predicts expected fatal crashes by mode at a census block group level; 2. functional-class fatal crash rates, used to disaggregate those block-group estimates onto individual roadway segments; and 3. severity ratios, used to extend the fatal crash estimates across the full range of crash severities. This model-based expectation is then refined using each segment's own crash record to produce a more locally accurate estimate. The sections below describe the Block Group model, the modelling process that links it to local observations, and the validation results. The modelling process is illustrated in Figure 3.

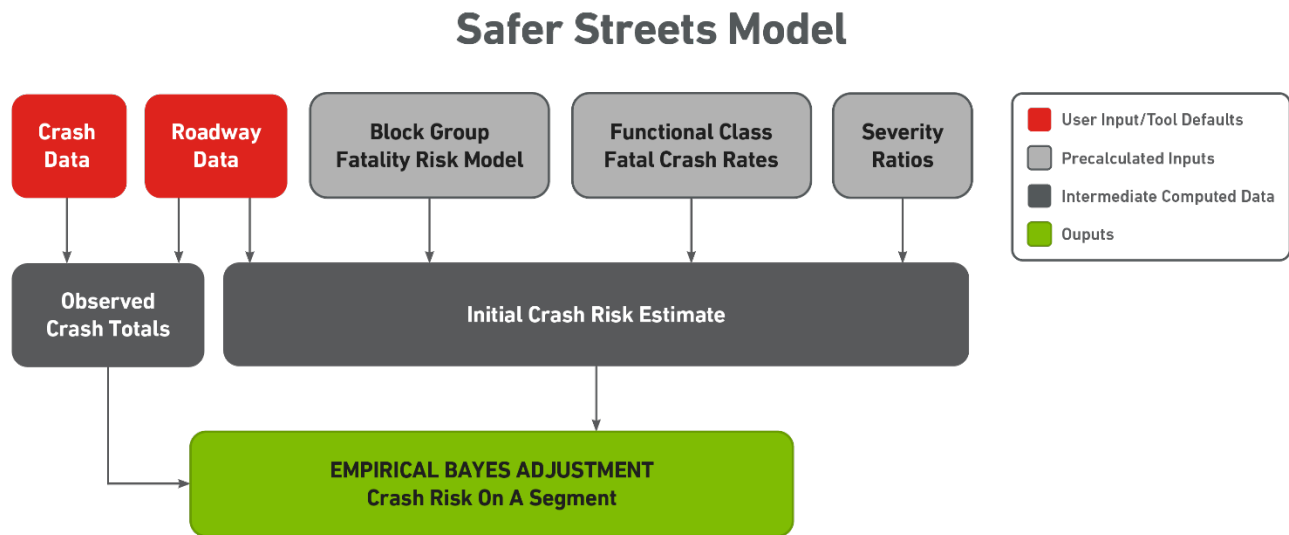


Figure 3: Safer Streets Model Process

3.2.1 BLOCK GROUP FATAL CRASH RISK MODEL

The previous version of SSPF incorporated a predictive model at the Census Tract level that was developed using fatality data from 2012 to 2016, and published in 2018. That model was estimated for pedestrians only, using separate models for urbanized and rural areas, and calibrated and applied to bicyclists within the tool post-hoc.

SSPF 2.0 is built on a new suite of statistical models. These models were developed using a similar methodology but refined to include all travel modes, more geographic precision, newer fatality data, and newer and more explanatory variables. Twelve separate models were estimated, covering three separate modes (pedestrian, bicyclist, and motorist-only fatal crashes) across a four-tier urban-rural gradient based on the EPA National Walkability Index⁷. The four tiers, ordered from most urbanized to most rural, are "Most Walkable," "Above Average Walkable," "Below Average Walkable," and "Least Walkable".

All twelve models were estimated using a similar structure, input database, and model building procedure. First, random forest models were developed to rank potential predictors based on their relative importance in predicting crashes. These rankings were then used in a backwards and forwards stepwise zero-inflated negative binomial regression procedure to find a set of explanatory variables for each mode and geography. Random effect and random parameter variables were incorporated to help explain variation in baseline fatality rates and in localized responses to different variables across the United States.

Dependent and explanatory variables were derived from five different public data sources: the Fatality Analysis Reporting System (FARS; fatal crash data), the Highway Performance Monitoring System

⁷ US EPA's National Walkability Index User Guide and Methodology - <https://www.epa.gov/smartgrowth/national-walkability-index-user-guide-and-methodology>

(roadway speeds, functional classification, and volumes), the Smart Location Database (intersection and segment density, population and household density, land use proxies derived from employment sector data, etc.), the Census Bureau American Community Survey (population demographics and socioeconomic status), and the Census Bureau Longitudinal Employer-Household Dynamics database (employment statistics). We also tested variables from the US Department of Transportation Equitable Transportation Communities database, though these were not ultimately retained in the final models.

Like the Tract-level model in the previous tool, these Block Group models are not perfect, but they are useful for identifying areas that may benefit from further investigation and safety treatments, including use in SSPF 2.0 for informing the Bayesian priors. Validation results show that the block groups in the top 20% by predicted fatality counts experience over 50% of the nation's fatal crashes. Out-of-sample testing shows that the models fit reasonably well. In the pedestrian model, the root mean square error (RMSE) within-sample was 0.36 compared to 0.37 out-of-sample. For bicyclists, the RMSE was 0.13 both within and out-of-sample. For motorists, the RMSE was 1.56 in-sample and 1.61 out-of-sample.

More details about the national Block Group Fatal Crash model can be found in the [full block group model report](#)⁸ available through the tool website.

3.2.2 MODELLING PROCESS

The model proceeds in four steps: 1. national block-group fatal crash estimates are disaggregated onto the roadway network; 2. disaggregated segment level fatal crash estimates are extended across the full KABCO severity scale; 3. local crash observations are redistributed along nearby segments; and 4. prior and observations are combined through a Bayesian update to produce the final segment-level estimate. The following sections provide an overview of each of these steps. The detailed process description and the mathematical specification are provided in the [model process document](#)⁹ available via the tool website.

3.2.2.1 DISAGGREGATE BLOCK-GROUP FATAL ESTIMATES ONTO ROADWAY SEGMENTS

National block-group fatal crash prediction models produce a fatal crash estimate for every Census block group in the United States for each mode. These estimates are disaggregated onto the roadway network by assigning each short segment to the block group or groups it overlaps, weighting by a buffered area-of-influence, and scaling so that segment-level estimates within each block group sum to the block-group total. Segments that border open water or other un-mapped land have their buffers re-normalized so that no fatal risk is lost.

3.2.2.2 EXTEND FATAL-ONLY ESTIMATES TO THE FULL KABCO SCALE

Block-group models estimate fatal (K) crashes only. To produce non-fatal estimates, each segment's fatal estimate is multiplied by state- and mode-specific severity ratios that give the typical number of A,

⁸ Block Group Fatal Crash Model Report - https://sspf.tooledesign.com/assets/documents/SSPF_v2_Block_Group_Model_Report.pdf

⁹ SSPF v2.0 Model Process - <https://sspf.tooledesign.com/assets/documents/Bayesian%20Model%20Updates%20Memo%20-%20v2.pdf>

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B, C, and O crashes per fatal crash. Severity ratios are pre-computed from publicly available state crash data. Where a state does not publish sufficient data, the tool falls back to a Census-region average (Northeast, South, Midwest, or West) shown in Figure 4.

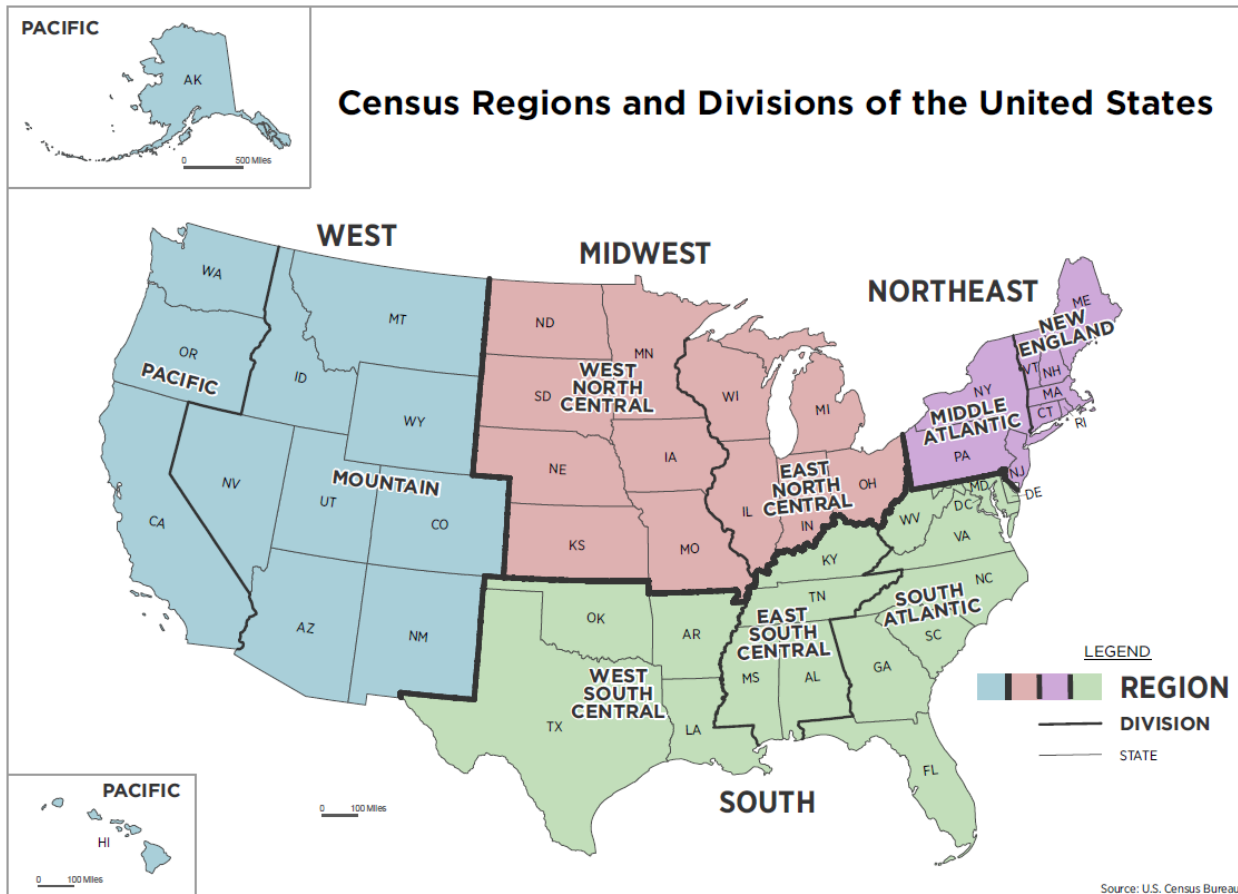


Figure 4: Census regions used for data gap filling (source: U.S Census Bureau)

3.2.2.3 REDISTRIBUTE OBSERVED CRASHES ALONG THE NETWORK

Each observed crash is joined to the roadway network and then distributed along nearby segments rather than assigned to a single point. The largest fraction of each crash stays on the segment where it occurred; smaller fractions are assigned to adjacent segments with weight decreasing linearly with distance from the crash. This step recognizes that recorded crash coordinates are imprecise and that a crash represents risk along a stretch of roadway, not just at a single point.

3.2.2.4 BAYESIAN UPDATE

The final estimate for each segment is a weighted blend of the model-based prior and the redistributed observed count. The weight given to the prior is a function of the prior's confidence - specifically, the coefficient of variation of the underlying block-group estimate. Where the confidence level for the prior estimates is high and observed crashes are sparse, the prior dominates; where observed crashes are

dense and the level of uncertainty is high for the prior, the observations dominate. Calibration of the blending parameters is described in Section 7.

Once segment-level crash counts by mode and severity are produced, they are multiplied by the configured crash costs to yield a societal cost estimate per segment. The cost-weighted output is what the dashboard, map, and report use to rank corridors for prioritization.

3.2.3 MODEL VALIDATION

The Safer Streets Model was validated through three case studies that span the range of community types in which the tool is expected to be applied: a dense urban context (City of New Orleans, LA), a small-to-mid-size suburban context (City of Albany, OR), and a low-density rural context (Knox County, OH). Detailed validation methodology is described in the [validation document](#)¹⁰ available from the tool website.

3.2.3.1 VALIDATION APPROACH

Each ten-year crash record was split in half. The first five years served as the input crash history available to the model - the data the tool would see at the time of analysis. The last five years served as a holdout - the future crash pattern the model is implicitly trying to predict. For each study area and each mode, three ranked lists of segments were produced: a baseline ranking from the observed crash history alone (the current state of practice in most agencies), a model-based ranking from the Safer Streets Model run on the same input period, and a target ranking from the holdout future crashes. The first two were then evaluated against the target.

Two ranking metrics were used at the top 10% and top 25% of segments: NDCG@k, which measures whether the segments with the most future crashes appear near the top of the ranking, and Cap@k, the percentile capture rate, which measures the share of future crashes that occur on segments ranked in the top k%. Both metrics are reported by mode and by case study. Top-k metrics are preferred over global rank correlations for a screening use case because what matters for prioritization is the quality of the top of the list, not the full ordering.

3.2.3.2 KEY FINDINGS

The Safer Streets Model improved ranking quality over the crash-history baseline most where crash history alone is least reliable. The largest gains appeared in the rural case study and for pedestrian and bicyclist modes. In the rural case, the Cap@25% metric for pedestrian crashes improved from 0.338 to 0.684 (+0.346), and for bicyclist crashes from 0.529 to 0.895 (+0.366). In the suburban case, pedestrian Cap@25% improved from 0.598 to 0.877 (+0.279) and bicyclist Cap@25% from 0.590 to 0.785 (+0.195). In the urban case, where crash history is dense and the baseline is already informative, gains were modest - pedestrian Cap@25% improved by 0.013 and bicyclist Cap@25% by 0.031.

Motor vehicle results followed a different pattern. In the rural case, the model improved motor vehicle Cap@25% from 0.695 to 0.741 (+0.046). In the suburban and urban cases, motor vehicle performance was essentially indistinguishable from the baseline - gains and losses on the order of a few

¹⁰ SSPF v2.0 Validation and Calibration Memo - https://sspf.tooledesign.com/assets/documents/Task_4_Model_Validation_Methodology_and_Results.pdf

thousandths. Motor vehicle crashes are sufficiently dense in suburban and urban networks that observed history is already a strong predictor of future patterns; the model contributes little additional improvement in those contexts.

These findings have a clear practical implication: the value added by the Safer Streets Model is largest precisely where conventional crash-history prioritization is weakest - in low-density networks and for vulnerable road user modes where crash data are sparse and statistically noisy. In contexts where crash history is dense and reliable, the model does not displace conventional methods; it confirms and complements them.

3.2.3.3 CALIBRATION

The Bayesian update has three tunable parameters: α , the maximum weight the prior can receive; β , the decay rate that determines how quickly the prior's weight drops as its coefficient of variation grows; and `blur_size`, the spatial smoothing radius used during observed-crash redistribution. A grid search across (α , β , `blur_size`) combinations was conducted on all three case studies and all three modes. Performance was averaged across modes and case studies to identify a balanced parameter set rather than tuning to any single context.

The selected calibrated values are $\alpha = 0.6$, $\beta = 3$, and `blur_size = 4`. Within the grid, model performance was relatively insensitive to α and β across reasonable ranges, with the exception of α values close to 1.0, which caused the prior to dominate observations and reduced performance. `blur_size` involved a trade-off: larger blur values smoothed year-to-year variation and improved ranking metrics, but at the cost of spatial precision in the final results. The selected value preserves enough spatial resolution to support corridor-level prioritization while reducing noise from year-to-year crash variation.

4. DATA INPUTS

SSPF v2.0 accepts three local input datasets - a study boundary, a roadway network, and a crash dataset. These are supplemented with six pre-processed datasets that are included with the tool. Users may run an analysis using only the pre-processed defaults, or they may substitute their own roadway and crash data where higher-quality local sources are available.

4.1 LOCAL INPUTS

4.1.1 STUDY BOUNDARY

The study boundary defines the geographic area to be analyzed. Users can specify the boundary in one of two ways. First, users may upload a custom boundary as a polygon-type shapefile. The shapefile must be provided as a single .zip archive containing the four required components - .shp, .shx, .dbf, and .prj. The .zip file cannot exceed 20 MB. Alternatively, users can select a pre-populated U.S. county or parish from dropdown menus for state and county/parish. No more than one county or parish may be selected per study. While the study boundary itself is not subject to a fixed area limit, the practical scale of an analysis is constrained by the size of the roadway network and the volume of crash records that can be processed. See the road network and crash data sections for the applicable limits.

4.1.2 ROAD NETWORK

The default roadway network is OpenStreetMap (OSM) centerlines, filtered to remove private and otherwise inaccessible links. Users may upload a local roadway network in the form of a shapefile. The shapefile must be provided as a single .zip archive containing the four required component files - .shp, .shx, .dbf, and .prj. The .zip file cannot exceed 20 MB. The total roadway length in the study area must not exceed 45,000 miles. This limit was set based on the OSM roadway mileage for the largest county we tested (Los Angeles County, CA). The shapefile must include the following attributes (the exact attribute names in the source data do not need to match the labels below):

- Unique ID - a unique value identifying each segment.
- Road Name - a text value.
- Functional Classification - a value that can be mapped to one of the standardized categories: Expressway, Major Arterial, Minor Arterial, Major Collector, Minor Collector, or Local Road.

As the user follows the tool's upload steps, they will be prompted to indicate which columns in their data store the unique ID, functional class, and road name attributes. They will then assign each functional class value in their data to one of the standardized categories listed above. This mapping step is necessary because functional class naming conventions vary across jurisdictions. Any limited-access highway should be mapped to Expressway. To omit a class of roads from the analysis (for example, Expressways), the user would assign those values to the 'Omit from Analysis' category. If the data does not have a clear match to a standardized class, users should assign it to the closest matching category; if necessary, users can review the initial analysis results and adjust the functional class assignments before re-running the analysis, so the results best reflect actual conditions.

4.1.3 CRASH DATA

The default crash dataset is the Fatality Analysis Reporting System (FARS), covering years 2018 to 2022. FARS contains only fatal crashes; analyses run with the default crash dataset will have a sparse sliding-windows result and Safer Streets Model output constrained to fatal severity. For most practical applications, a local crash dataset covering the most recent five years of all-severity crashes is strongly recommended.

Local crash data must be uploaded as a point-type shapefile. The shapefile must be provided as a single .zip archive containing the four required components (.shp, .shx, .dbf, and .prj), and the .zip file cannot exceed 20 MB. The total number of crashes must not exceed 250,000. The shapefile must include the following attributes and characteristics (the exact attribute names in the source data do not need to match the labels below):

- **Report ID:** a unique ID for each crash, ensuring one record per crash. If the dataset has multiple records per crash, the user may be working with a person, unit, or vehicle table. In that case, the user need to restructure their data, or ask their crash data provider for help obtaining a dataset with one row per crash, before uploading.
- **Crash Year:** a four-digit integer. The tool is designed to use five years of crash data. If fewer years are available, the analysis can still run with the available years, but results may be less reliable.
- **Severity:** must be mapped to the KABCO scale as Fatality (K), Incapacitating Injury (A), Non-Incapacitating Injury (B), Possible Injury (C), or Property Damage Only (O). Users may also assign values to 'Omit From Analysis' to exclude those crashes. Severity should represent the most severely injured person or the worst outcome associated with the crash.
- **Crash Mode:** each crash must be assigned to bicycle, pedestrian, or motor vehicle mode. Mode identification must occur in a single column; if data uses separate columns to indicate bicyclist or pedestrian involvement, it needs to be restructured into a single mode column before uploading. For crashes involving multiple modes, the most common conventions are to assign the entire crash to the mode of the most severely injured person, or to the mode of the most vulnerable road user (typically in the order pedestrian, bicycle, motor vehicle). Users may also assign values to 'Omit From Analysis' to exclude those crashes.

Crash datasets with more than 50,000 records may lead to slower map performance in the app; the analysis can still run, and outputs can still be downloaded for use in other GIS tools. If the crash data contains only fatal crashes (for example, FARS), non-fatal model results and non-fatal crash costs are not calculated.

4.2 BUNDLED TOOL DATASETS

Six national datasets are pre-computed and bundled with the tool so that an analysis can be run anywhere in the United States without additional data preparation. The block-group fatality model outputs provide the systemic prior for each mode, together with standard errors used to derive the coefficient of variation in the Bayesian update. The state-by-functional-class fatal crash rate table provides the default fatal-crash rate per mile of roadway per year, computed from five years (2018-2022) of FARS records and Highway Performance Monitoring System (HPMS) roadway mileage. The

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state-by-mode severity ratio table provides A, B, C, and O crash counts per fatal crash by state and mode, derived from publicly available state crash data with a Census-region fallback for states without sufficient data. Census state and county boundaries are sourced from the U.S. Census Bureau's TIGER/Line files. OpenStreetMap centerlines provide the default roadway network. FARS provides the default crash dataset. The [Resources page](#)¹¹ in the tool links to these datasets directly or provides a link to the script used to upload these datasets to the tool database.

¹¹ SSPF v2.0 Resources Page - <https://sspf.tooledesign.com/resources>

5. ANALYSIS SETTINGS

Most analysis parameters in SSPF v2.0 can be reviewed and adjusted before an analysis is run. Each parameter has a default selected to be reasonable for most U.S. study areas; the defaults can be overridden where local policy, local crash characteristics, or specific analytical questions warrant. These user configurable parameters fall into four groups: severity weights, window lengths, crash costs, and other settings. Their defaults and allowed ranges are summarized in Table 1.

Table 1. Default values and allowed ranges for user-configurable analysis parameters in SSPF v2.0.

Parameter Group	Parameter	Default	Allowed Range
Severity Weights	Fatality (K)	3	0–100, integer; $K \geq A \geq B \geq C \geq O$
Severity Weights	Serious Injury (A)	3	0–100, integer
Severity Weights	Non-Serious Injury (B)	1	0–100, integer
Severity Weights	Possible Injury (C)	0	0–100, integer
Severity Weights	No Injury (O)	0	0–100, integer
Window Lengths	Sliding Window Length (mi)	0.5	0.25–2.0, in 0.05 mi increments
Window Lengths	Sliding-to-Short Window Ratio	5 (short window = 0.1 mi)	1–10, integer
Crash Costs (USD)	Fatality (K)	\$15,988,000	≥ 0, USD; $K \geq A \geq B \geq C \geq O$
Crash Costs (USD)	Serious Injury (A)	\$1,705,100	≥ 0, USD; $K \geq A \geq B \geq C \geq O$
Crash Costs (USD)	Non-Serious Injury (B)	\$384,000	≥ 0, USD; $K \geq A \geq B \geq C \geq O$
Crash Costs (USD)	Possible Injury (C)	\$204,600	≥ 0, USD; $K \geq A \geq B \geq C \geq O$
Crash Costs (USD)	No Injury (O)	\$18,100	≥ 0, USD; $K \geq A \geq B \geq C \geq O$
Other Settings	Crash Join Distance (ft)	50	25–250 ft
Other Settings	Number of Streets in Report	10	5–20, integer

5.1 SEVERITY WEIGHTS

Severity weights control how much each KABCO injury level contributes to the severity-weighted crash density score in the Sliding Windows Analysis. Weights must be integers between 0 and 100. The default weighting (3, 3, 1, 0, 0) reflects a common practice of treating fatal and serious injury crashes as

roughly equivalent for prioritization purposes, treating non-serious injury crashes as meaningful but less critical, and excluding possible injury and property-damage-only crashes from the density score. Agencies that wish to weight fatalities more heavily than serious injuries - for example, in a Vision Zero context with greater importance on fatal and severe injury crashes - can adjust the K and A weight upward. Ordinality is enforced: $K \geq A \geq B \geq C \geq O$.

5.2 WINDOW LENGTHS

Window lengths are governed by two parameters: the sliding window length and the sliding-to-short-window length ratio. The sliding window length controls how long a corridor must be to register as a single high-crash segment. The default of 0.5 miles balances corridor-level interpretability against spatial resolution. Longer windows produce smoother, more aggregated results that emphasize multi-block corridors; shorter windows produce sharper results that emphasize individual intersections and short segments. The sliding-to-short-window length ratio controls how finely the sliding window advances along the network, with a default ratio of 5 producing a short underlying segment of 0.1 miles. These short windows serve as the unit of analysis for both the sliding windows analysis and the Safer Streets model. Studies with low-density context (e.g., rural areas) may want to use longer window lengths. Users may start with the defaults, review results and adjust iteratively as needed.

5.3 CRASH COSTS

Crash costs translate the modeled segment-level crash counts into societal cost estimates that can be used for benefit-cost analysis, project prioritization, and stakeholder communication. The defaults are the comprehensive societal costs published by FHWA¹² and reflect the broad set of costs imposed by traffic crashes, including medical, emergency response, lost productivity, property damage, and quality-of-life impacts. Agencies that use a different cost reference (for example, a state-specific cost value of life) can substitute their own values. Ordinality is enforced: $K \geq A \geq B \geq C \geq O$.

5.4 OTHER SETTINGS

The crash join distance determines how far from a roadway segment a crash point may lie and still be associated with that segment during preprocessing. The default of 50 feet handles the vast majority of crash points without sweeping in crashes that occurred on parallel facilities. The number of streets in the report controls how many top-ranking corridors are highlighted in the compiled PDF report; agencies producing a focused investment list typically use the default of 10, while those producing a longer planning document may extend it to 20.

¹² FHWA's Crash Costs for Highway Safety Analysis (2024) - https://highways.dot.gov/sites/fhwa.dot.gov/files/2025-10/CrashCostFactSheet_508_OCT2025.pdf

6. OUTPUTS

SSPF v2.0 presents results through four complementary outputs: 1. an interactive dashboard, 2. map visualization, 3. a compiled PDF report, and 4. a downloadable GIS data package. The first two are intended for exploration on the application platform; the latter two are intended for communication and for integration into downstream workflows.

6.1 INTERACTIVE DASHBOARD

The dashboard summarizes the analysis through an interactive map of the highest-scoring crash corridors paired with a series of charts describing the underlying crash patterns - severity breakdowns, modal split, functional class crash distribution, trends over time, model fit charts, and highest crash density corridors. The charts on the page are interactive, allowing the user to apply additional filters and export both the underlying tables and plots. The dashboard is the fastest way to develop an initial understanding of what the analysis is showing.



Figure 5: Dashboard page screenshot

6.2 MAP VISUALIZATION

The map visualization page allows any combination of crashes, segment outputs, and block group polygon data to be viewed on a map within the browser window. The map provides an interactive interface for users to explore their study area outputs and inputs all within the application. Users can click on different elements on the map to view more details. Note that sliding windows analysis and safer streets model results can also be visualized on the run analysis page of the tool. The run analysis page however, does not allow overlaying with crashes and block group model results which is only possible on the map visualization page.

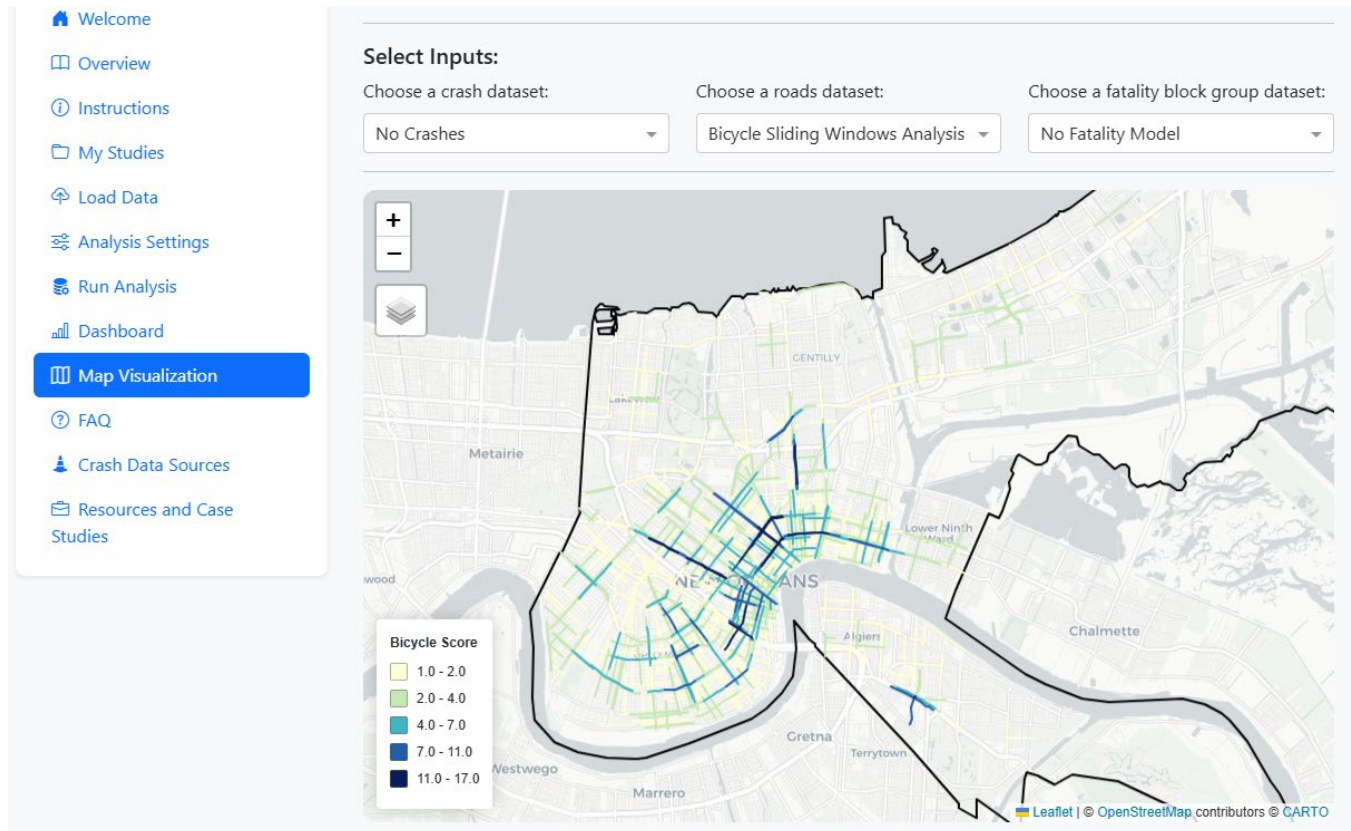


Figure 6: Map visualization page screenshot

6.3 COMPILED PDF REPORT

The compiled PDF report summarizes the study inputs, analysis settings, and key results in a format suitable for sharing with stakeholders. It includes sliding windows analysis and safer street model output maps, dashboard charts, highest crash density corridors, and additional information for the study. The PDF report is essentially a static document summarizing the study. If a study does not have a specific analysis complete or has certain data missing, the report will not have any content populated for those sections. The PDF report can be compiled and downloaded from the dashboard page of the tool.

6.4 GIS DATA PACKAGE

The GIS data package contains all datasets produced or used during the analysis: the segmented roadway network with the modeled crash estimates by mode and severity, model results as crash cost, and the sliding windows density results. It also has the study's input data and the underlying block group fatal crash model for the study area. The package is intended for users who will carry SSPF results into their own GIS environment for further analysis, mapping, or integration with project pipelines. The GIS data package can be compiled and downloaded from the dashboard page of the tool. The GIS data package comes in the form of a GeoPackage (GPKG) along with a data dictionary describing the attributes for the layers included.

7. DISCUSSION

SSPF v2.0 is best understood as an entry point to systemic multimodal safety analysis. It is designed for planners and engineers at municipal, county, regional, and state agencies who need to identify candidate corridors for safety investments but who may not have the GIS or statistical resources to develop a full systemic analysis from scratch. The tool is appropriate for early-stage screening, project pipeline development, Safety Action Plan supporting analysis, and quick-turn risk overviews. It is not appropriate as a substitute for detailed engineering studies or more nuanced safety analyses such as Safety Performance Functions (SPFs).

Several patterns are worth noting when reviewing SSPF output. The Safer Streets Model and the Sliding Windows Analysis answer different questions and will not always agree. Corridors that rank high on both are the strongest investment candidates. Corridors that rank high only on the model side are systemic opportunities - locations where conditions for severe crashes may exist but observed crash history is sparse or under-reported. These are precisely the locations that conventional crash-history methods may miss and that the Safer Streets Model is designed to identify.

Modeled crash counts are statistical estimates, not predictions of when or where a specific crash will occur. They should be interpreted as relative risk signals across the network - useful for ranking and prioritization, not as absolute forecasts. The same is true of the societal cost estimates derived from them: the cost values are most informative when compared across corridors within a single analysis, less informative as standalone numbers.

Divided roadways merit specific attention. When opposing directions of a divided facility are represented as separate centerlines in the input roadway data, the crash-redistribution step assigns observed crashes to the nearest centerline. Corridors with heavy directional separation may therefore show asymmetric results between the two directions. Where this is misleading, the input roadway network can be pre-processed to collapse divided centerlines to a single representative line before uploading.

Results based on the FARS default dataset are limited to fatal crashes. The sliding windows screen will appear sparse - most corridors will record zero or one severity weighted observation - and the Safer Streets Model output will be restricted to the fatal severity level. For this reason, local crash data, if available, is recommended over the default FARS data.

7.1 CAVEATS AND LIMITATIONS

The Safer Streets Model relies on broad-stroke inputs and assumptions rather than the granular roadway attributes - lane configuration, posted speed, intersection control, presence of bicycle facilities, lighting condition, sight distance, and other contextual information. SSPF produces relative risk signals across a network; it does not produce site-specific crash predictions, and it does not replace the engineering judgment required to design specific countermeasures at a specific location. A further limitation stems from the scale at which the systemic prior is estimated. The block-group fatality models that provide the systemic prior are estimated at the block-group scale; the disaggregation step

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distributes those estimates across segments within a block group based on functional class and segment length but cannot resolve within-block-group variation driven by location-specific design.

Where SSPF results identify a corridor or set of corridors that will receive substantial investment, the screening output should be followed by a more detailed analysis. That detailed analysis may include a calibrated SPF for the relevant facility type, a road safety audit, a corridor-level crash diagnosis with countermeasure selection, or a benefit-cost analysis using site-specific cost estimates. SSPF is intended to identify where more detailed work should be focused, not to substitute for it.

8. RESOURCES

SSPF v2.0 is freely available as an open-source web application. The tool is hosted at <https://sspf.tooledesign.com/>, the previous version 1.0 website URL - <https://www.saferstreetspriorityfinder.com/> - now redirects to the updated version 2.0 website. Methodology documentation, and case study materials are available at <https://sspf.tooledesign.com/resources>. The underlying codebase is in the tool's GitHub Repository - <https://github.com/tooledesign/Safer-Streets-Priority-Finder/>. Questions about the methodology, data, or application of the tool can be directed to the development team at saferstreetspriorityfinder@tooledesign.com.